Cole Durham

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Education

University of Connecticut Storrs, CT
Ph.D., Mathematics *Expected May 2027*
Advisor: Ovidiu Munteanu

University of Rochester Rochester, NY
Honors B.S., Mathematics; B.A., Physics *May 2021*

Research and Publications

Research Interests
Comparison geometry, geometric analysis, PDE.

Publications

- Cole Durham, "Bottom Spectrum Estimate Under Curvature Integrability Condition," *Journal of Functional Analysis* 290 (2026), No. 4, Paper No. 111254.

Teaching Experience

University of Connecticut Storrs, CT
Instructor of Record *Spring 2024; Spring 2025–Fall 2025*

- Independently taught Elementary Differential Equations and Business and Economics Calculus, with approximately 60–80 undergraduates per course.
- Designed course materials, including syllabi, assignments, and interactive Python visualizations; balanced mathematical rigor with accessible explanations and applications.

University of Connecticut Storrs, CT
Teaching Assistant *Fall 2021–Fall 2026*

- Supported Calculus I–III and Elementary Differential Equations through leading multiple discussion sections, explaining core concepts, and holding office hours.
- Served as Lead Teaching Assistant for Multivariable Calculus, coordinating with course instructors, managing TA schedules for Proctoring Center coverage and shared office hours.

University of Rochester Rochester, NY
Undergraduate Teaching Assistant *Fall 2020-Spring 2021*

- Assisted with Multivariable Calculus and Honors Abstract Algebra through office hours and grading of assignments.

Selected Projects

Data Science Demos for Business Calculus Python, Streamlit, pandas, Matplotlib

- Developed interactive, business-context demonstrations of introductory data science and machine learning concepts for business and economics students.

Honors Multivariable Calculus Walk-Throughs HTML

- Slideshow written in HTML to introduce probability concepts adjacent to calculus, allowing for visuals and simulations.

Honors and Awards

- Bausch & Lomb Science Scholarship
- Sigma Pi Sigma, Physics Honor Society (University of Rochester chapter)

Skills

Mathematics: geometric analysis, differential equations, multivariable calculus, linear algebra
Computing: Python, R, SQL, LaTeX, Git, Jupyter, VS Code, RStudio, Streamlit
Data analysis: regression, classification, time series, feature selection, cross-validation, statistical inference